Serambi.ai



Scaffold, Bukan Oracle

Belajar mandiri, tapi nggak sendirian.

Adaptive multi-agent learning assistant untuk career switcher Indonesia

IBM SkillsBuild University Education National Hackathon

M. Muchsin

Belajar mandiri — dan banyak yang gagal di tengah jalan



Tidak ada roadmap personal

Materi acak dari YouTube, Udemy, blog — tanpa peta ke target role.

Progres tidak terlacak

Tidak ada jawaban: sudah sejauh mana, lanjut ke apa?

AI generik memicu cognitive offloading

Langsung dijawab — belajar aktif tidak terjadi (BJET, 694× dikutip).

Alternatif tidak terjangkau

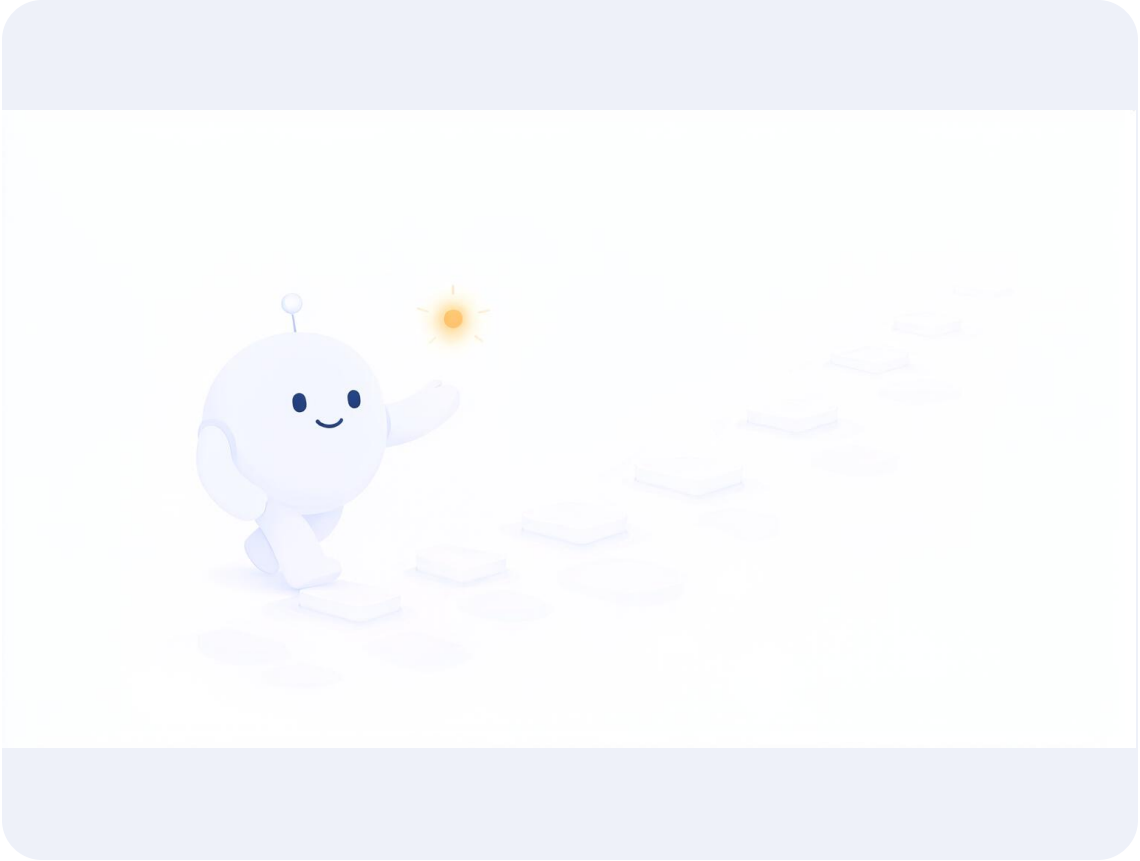
Bootcamp & mentor privat mahal; kursus online one-size-fits-all.

Kesenjangan pasar: upskilling global tumbuh CAGR 11–14% — hampir seluruhnya B2B/korporasi; career switcher individu masih tidak terlayani.

Scaffold, Bukan Oracle

Serambi.ai memandu learner ke jawaban — tidak menjawab untuk mereka.

1. **Ketik goal**
   “Guru mau switch ke AI engineer”
2. **Roadmap mingguan**
   Goal Agent memecah goal jadi peta
3. **Feynman tutoring**
   Tutor Agent memancing, bukan menjawab
4. **Kuis adaptif**
   Assessment Agent + graduated hints
5. **Spaced repetition**
   Review sesuai forgetting curve pribadi
6. **Pantau wellbeing**
   Wellbeing Agent: motivasi, cegah burnout



Anchor pedagogis: GenMentor (FWCI 47.3) · IntelliCode (FWCI 23.5) · Feynman Bot (arXiv 2506.09055) · Spaced Repetition

Career switcher Indonesia: 22-35, non-tech, belajar mandiri

Persona primer

Profesional 22–35, background non-tech

Guru, admin, marketing — mau pivot ke
AI engineer, data analyst, backend dev
belajar mandiri, 1–6 bulan.

Waktu terbatas | Bahasa Indonesia | Anggaran terbatas

Beachhead: individu yang tidak terlayani pasar B2B

Pengguna sekunder

Tech worker naik level / pivot domain

Pasar & willingness-to-pay

OECD 2024: kesenjangan tech skills melebar;
upskilling CAGR 11–14% fokus B2B (studi Politecnico)

Revenue (skenario)

Skenario

Freemium: roadmap gratis · pro = kuis + spaced repetition
B2B whitelabel: bootcamp, kampus, program reskilling

"Saya guru, mau pakai AI untuk switch ke AI engineer — sudah coba-coba, tapi tidak tahu mulai dari mana."

— Job-to-be-done, submission draft

3 langkah: goal → roadmap → belajar terpandu

1

Ketik goal

“Guru mau switch ke AI engineer”

2

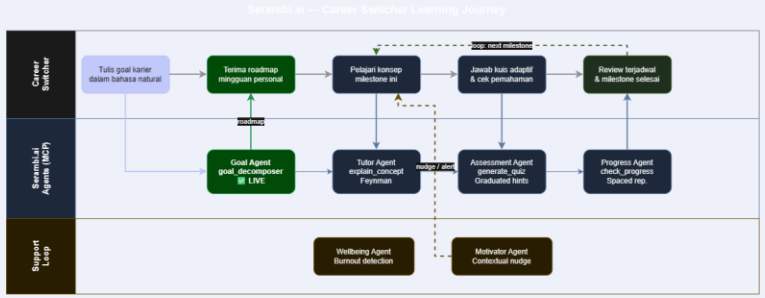
Dapat roadmap mingguan

Deadline, milestones, first action — < 1 menit

3

Belajar terpandu

6 agent: Feynman, kuis adaptif, spaced repetition, wellbeing



E2E verified: Bob → MCP goal_decomposer → JSON roadmap 12 minggu

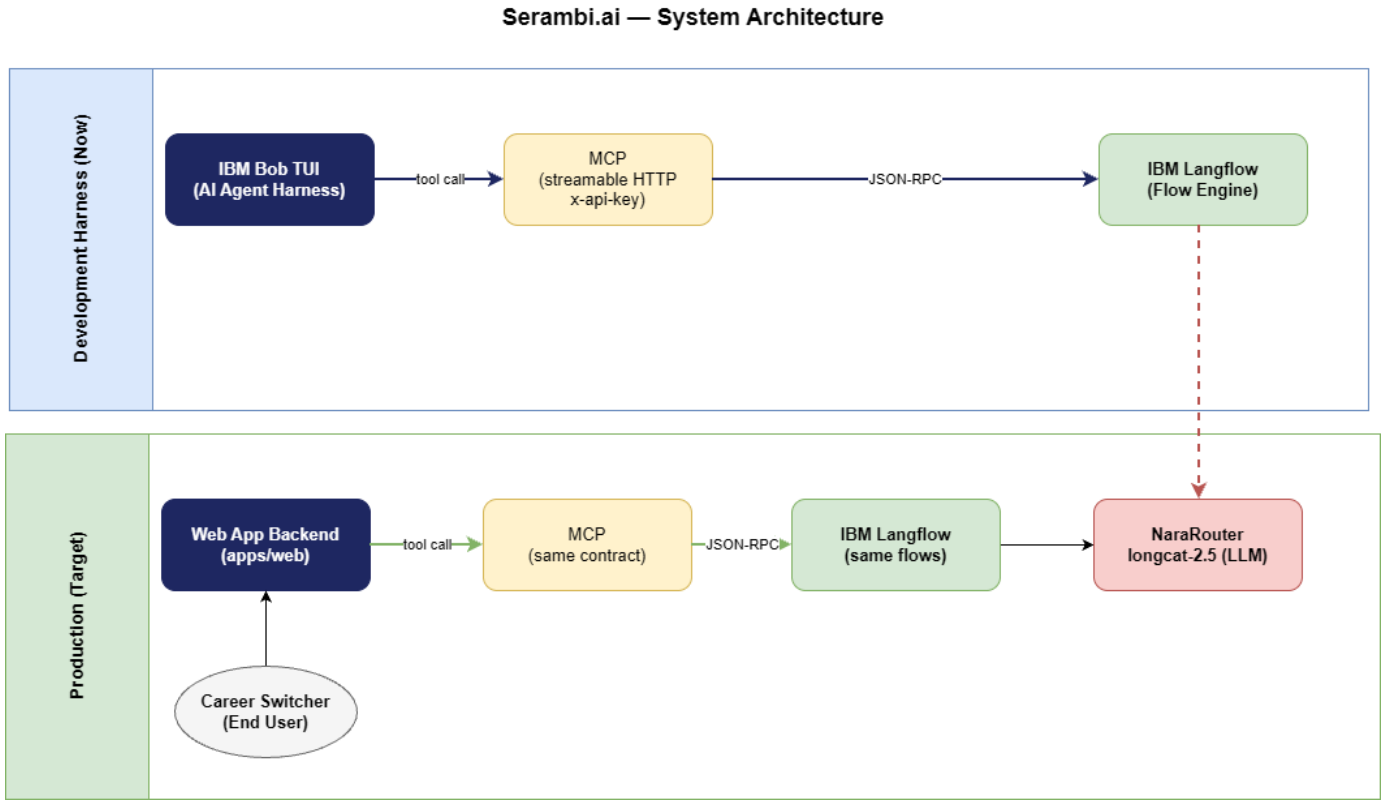
IBM Bob × Langflow × MCP — kontrak yang sama di dev dan produksi

IBM Bob — agent harness
build, test, verifikasi agent e2e

IBM Langflow — flow engine
tiap flow = 1 MCP tool

MCP — streamable HTTP
bridge Bob ↔ Langflow; kontrak produksi

NaraRouter / longcat-2.5
LLM backbone, temperature 0.3



Anchor riset: GenMentor (FWCI 47.3) · IntelliCode (FWCI 23.5) · Feynman Bot (arXiv 2506.09055) · Zaidi et al. (arXiv 2004.11327) · BJET (694× dikutip)

6 agent, 5 fitur — 1 sudah live e2e

Goal Decomposer
LIVE

goal → roadmap JSON mingguan (MCP tool goal_decomposer)

Tutor (Feynman)

probing questions — AI memancing refleksi, bukan menjawab

Assessment (hinting)

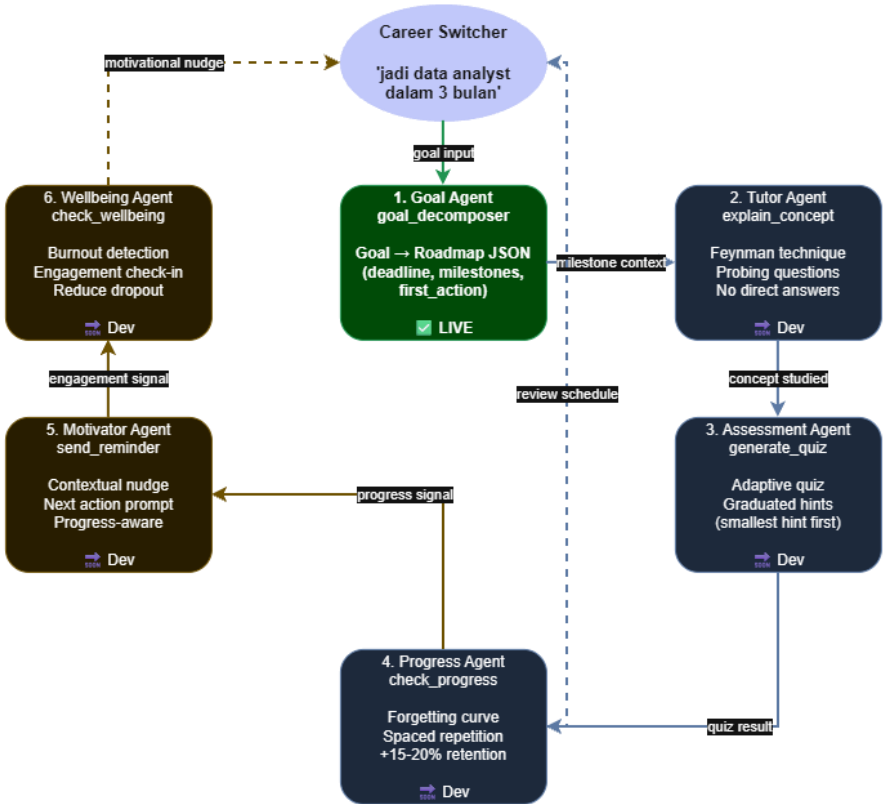
kuis adaptif; salah → hint terkecil dulu (IntelliCode)

Progress (spaced rep)

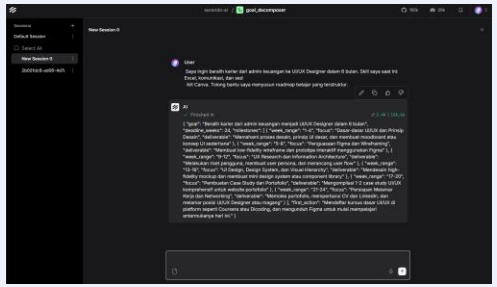
jadwal review per forgetting curve — +15–20% retensi (Zaidi et al.)

Motivator & Wellbeing

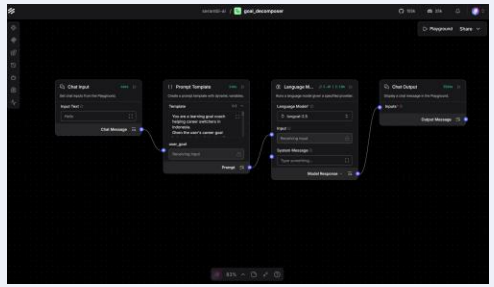
nudge kontekstual + cek burnout berkala



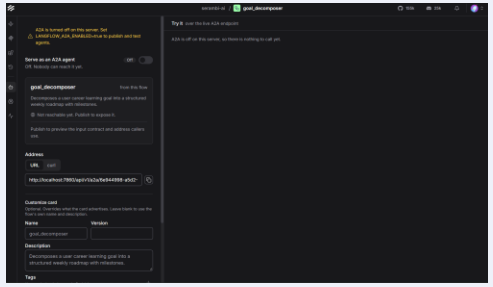
Bukti, bukan janji: agent Langflow benar-benar dieksekusi



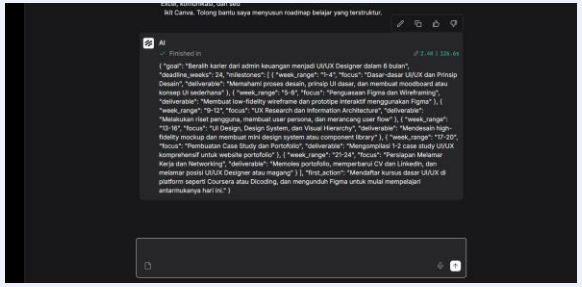
[01] Bob TUI — harness dev memanggil goal_decomposer



[02] Canvas Langflow — flow goal_decomposer



[03] Endpoint Langflow — nama flow + deskripsi agent



[04] Output nyata — JSON roadmap 12 minggu

Bukan dijawab LLM Bob — agent Langflow yang dieksekusi via MCP

Dampak terukur, AI yang bertanggung jawab

≥ 15% — Skenario
Career Goal Progress Rate per minggu (North Star — target)

< 1 menit
Roadmap personal — dari 2–4 jam riset (bukti demo e2e)

+15-20%
Retensi vs spaced repetition tetap (Zaidi et al.)

g = 1.613
Meta-analisis AI & self-regulated learning (32 studi)

80%+
Peserta memilih Feynman vs membaca pasif (arXiv 2506.09055)

1.000 — Skenario
Pengguna aktif per bulan pada bulan ke-6 (target)

Privasi
data progres milik user, tanpa penjualan PII

Human-in-loop
scaffold memaksa berpikir (counter offloading, BJET)

Wellbeing
deteksi burnout + rujukan manusia, bukan nasihat medis

Transparansi
setiap pilihan pedagogis tertaut paper terpublikasi

MVP live — 5 Agent + web app dalam 3-6 bulan

Selesai

- [x] Goal Agent e2e verified
- [x] Infra MCP + Langflow + Bob
- [x] Dokumentasi + submission draft
- [x] Verification scripts (flow + e2e)

3–6 bulan

- [ ] Tutor (Feynman)
- [ ] Assessment (hinting)
- [ ] Progress (spaced rep)
- [ ] Motivator + Wellbeing
- [ ] Web App
- [ ] WhatsApp bot surface

B2B roadmap
Whitelabel bootcamp & kampus

Skala teknis
Langflow Cloud multi-tenant

Moat
Forgetting curve per domain

Dibangun berdasarkan riset ilmiah

Riset mendalam

30+ paper (abstract + DOI);
topik S2: LLM hallucination

Implementasi nyata

Goal Agent e2e verified,
bukan mockup

Full-stack & agentic

Paham stack produksi
sampai infra agent

Pedagogi berbasis bukti

Tiap fitur tertaut paper
ber-FWCI tinggi

Belajar mandiri, tapi nggak sendirian.

Silakan cek langsung: github.com/mmuchsin/serambi-ai